A REGULAR LOCAL RING OF CHARACTERISTIC P WITH FINITE DEGREE KAPPA OVER KAPPA^P IS EXCELLENT IFF ITS COMPLETION MAP IS FORMALLY ETALE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The forward direction delegates the formal smoothness to Neron-Popescu desingularization (Smoothing Ring Maps, Theorem smoothing-theorem-popescu) - load-bearing but named; content verbatim, no rewrite.

Editorial note. The label lemma-excellent-regular-local-rings names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Regular local rings of characteristic p, the Frobenius Twist [kappa : kappa^p], and excellence
- Formal etaleness = formal unramifiedness plus formal smoothness; Omega_{A^/A} = 0 detects formal unramifiedness
- The Neron-Popescu desingularization (Smoothing Ring Maps, theorem-popescu): the completion of an excellent G-ring is a filtered colimit of smooth A-algebras; NL and the vanishing criterion for formal smoothness (Algebra, proposition-characterize-formally-smooth)

Source reference. Original proposition 031J: sources/stacks-chow/tags.txt, source lines 3986–3986. Prerequisite pointer invoked at 3602; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0G66: sources/stacks-examples/pinned-original-source.tex, source lines 3484–3484. Prerequisite pointer invoked at 3590; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 07GC: sources/stacks-chow/tags.txt, source lines 9512–9512. Prerequisite pointer invoked at 3599; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a regular local ring of characteristic $p > 0$. Suppose $[\kappa : \kappa^p] < \infty$. Then A is excellent if and only if $A \to A^\wedge$ is formally étale.*

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The backward implication follows from Lemma 0G66. For the forward implication, note that we already know from Lemma 0G66 that $A \to A^\wedge$ is formally unramified or equivalently that $\Omega_{A^\wedge/A}$ is zero. Thus, it suffices to show that the completion map is formally smooth when A is excellent. By Néron-Popescu desingularization $A \to A^\wedge$ can be written as a filtered colimit of smooth A-algebras (Smoothing Ring Maps, Theorem 07GC). Hence $NL_{A^\wedge/A}$ has vanishing cohomology in degree -1. Thus $A \to A^\wedge$ is formally smooth by Algebra, Proposition 031J. □